puntr.

PRODUCT AND TECHNICAL BLUEPRINT

Puntr Football Intelligence Engine.

Verified football data for Puntr: AI extracts, code validates, independent sources corroborate.

VERSION 1.0	DATE 5 November 2026	STATUS Draft for owner review
TECHNICAL COMPLETION 9 November 2026	PUBLIC LAUNCH 1 December 2026	SITE puntr.co.za

Fixtures and results only. No live scores or betting odds. 18+ · Gamble responsibly: 0800 006 008

PUNTR FOOTBALL INTELLIGENCE ENGINE

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Version **1.0** Date **5 November 2026** Status **Draft for owner review**

> **How to read this document.** Every claim about a provider is tagged. VERIFIED means verified from the provider's own page on 5 November 2026. PROPOSED means proposed: our design, not yet built. CONFIRM means to confirm at implementation: the provider's page did not settle it. Nothing tagged PROPOSED or CONFIRM is a promise, and no data-licensing permission is assumed anywhere.

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1 Executive summary

PFIE is Puntr's own football data layer. It collects fixtures and results from several sources, checks them with ordinary code, stores them with their source and a verification stamp, and serves them to the Puntr app, to CSV and JSON feeds, and later to a public API.

Three design decisions from the owner, now built into this blueprint:

1. **AI is not responsible for data truth.** Gemini extracts. Deterministic application code validates. Independent sources corroborate. A model never decides which fixture is correct (section 7).
2. **Founding-member acquisition comes first.** The early-access landing page, sitewide copy and sign-up forms ship before PFIE, while PFIE is still being built. Status: built and on branch `mvp/launch` in pull request #42, waiting for the owner to merge it (section 10, Phase 0).
3. **PFIE is a proprietary platform that becomes an API.** The internal model is API-first, so that Puntr can later supply sports prediction data to other platforms (sections 2 and 13).

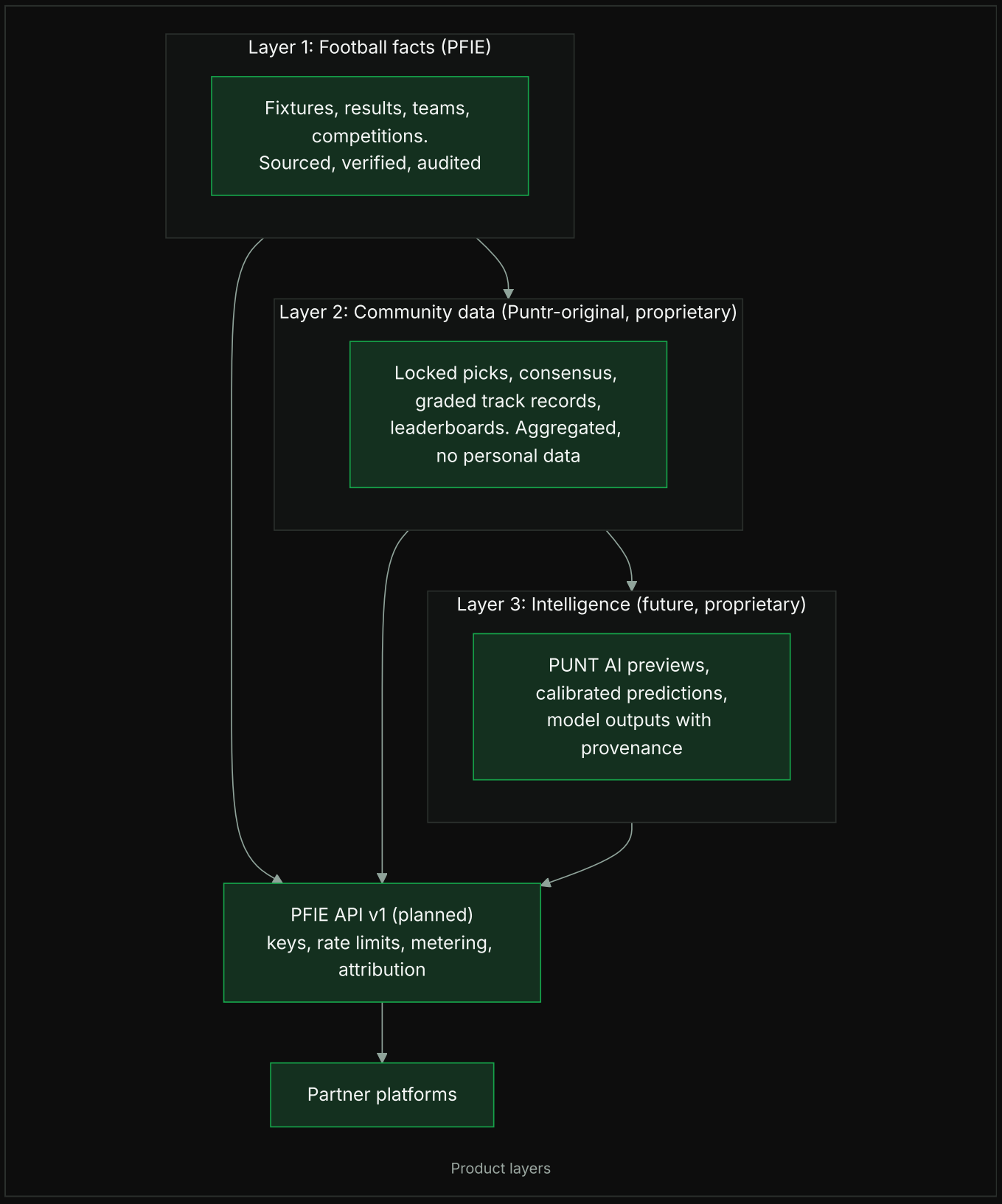
What the research found (all verified 5 November 2026):

Finding	Why it matters
Firecrawl Free is **1,000 credits a month**, 10 requests a minute on scrape and search, 2 concurrent browsers `VERIFIED`.	About 33 pages a day. The six-hourly refresh of 50 competitions in the brief would need roughly 6,000 credits a month. The schedule must be tiered (section 8).
Firecrawl's JSON extraction mode costs **4 extra credits** a page `VERIFIED`.	We fetch markdown only (1 credit) and extract with Gemini.
Gemini's free tier states **"Used to improve our products: Yes"** `VERIFIED`.	Only public, non-personal fixture pages may ever be sent to Gemini. This matches the project rule that personal data never goes to an AI service.
Parallel gives **$5 of credit a month**, about 5,000 searches, and says a payment card activates it and extra usage is billed to the card `VERIFIED`.	Not truly R0 unless the card cannot be charged. Owner decision needed (section 8).
Vercel Hobby is restricted to non-commercial use and cron runs **once a day** `VERIFIED`.	Puntr carries affiliate and advertising units, which Vercel's definition of commercial use covers. If the project is on Hobby, this is a terms risk that PFIE does not cause but cannot ignore. Scheduling uses Supabase pg_cron, as the repository already does.
Supabase Free has **no automatic backups**, 500 MB of database and 500,000 edge invocations a month `VERIFIED`.	PFIE needs its own backup job (section 9).
Fixture data collected from third parties may carry database or copyright protection, and sites' terms apply `CONFIRM`.	The proprietary asset worth building an API around is Puntr's own community and prediction data, not resold third-party fixtures (section 13). `ATTORNEY REVIEW`

Scope honesty on the 9 November date. It is four days away. A credible technical completion by then is the core platform: schema, competition registry, a public-domain importer, the extraction and validation pipeline for the priority competitions, corroboration, an admin review queue, CSV and JSON feeds, and monitoring. Reaching all 50 competitions with corroborated data is a rolling activity from 9 November to launch on 1 December, competition by competition. The technical deadline and the launch date are kept separate throughout.

2 Product vision

PFIE is the foundation of three layers. Only the first is being built now.



Product layers

- **Layer 1, football facts (PFIE).** Fixtures, results, teams, competitions. Every record carries its sources, a verification state and an audit trail.
- **Layer 2, community data (Puntr-original).** Locked picks, graded track records, consensus and leaderboards. This is generated by Puntr's members and owned by Puntr. It is the strongest candidate for a licensable data product and the one that fits "sports prediction data" best.
- **Layer 3, intelligence (future).** PUNT AI previews and calibrated predictions, always with provenance.

What PFIE is not at MVP. It does not provide real-time live scores, betting odds or bookmaker data, and it does not claim to. Fixture collection is a scheduled process with a "last verified" time on every record. Odds stay out of scope in line with the launch rules.

3 Functional requirements

Priority key: **M** must for 9 November, **S** should by launch on 1 December, **C** could after launch.

ID	Requirement	Pri
FR-1	Discover upcoming fixtures across the competition registry	M
FR-2	Capture home and away team, competition, season, round, date, kick-off, venue	M
FR-3	Detect and record postponed, cancelled and rescheduled fixtures	M
FR-4	Extract structured data with Gemini using a fixed JSON schema and evidence quotes	M
FR-5	Validate every extraction with deterministic code before it can be stored	M
FR-6	Normalise team and competition names to canonical IDs (alias tables)	M
FR-7	Detect duplicates with one natural key per fixture	M
FR-8	Corroborate across independent sources and record every observation	M
FR-9	Compute a confidence score by a fixed formula, never by the model	M
FR-10	Keep source, source URL and verification timestamp on every record	M
FR-11	Collect completed results and store full-time scores	M
FR-12	Never overwrite a verified result with incomplete data; corrections go through review and are logged	M
FR-13	Keep a central registry of competitions, teams, fixtures, results, sources, runs, errors and changes	M
FR-14	Serve the Puntr app through the existing `public.fixtures` table	M
FR-15	CSV and JSON exports	M
FR-16	Admin queue for disputed fixtures and unresolved team names	M
FR-17	Quota ledger, dynamic schedule reduction and exhaustion handling	M
FR-18	Daily historical reconciliation and weekly integrity audit	S
FR-19	Source reliability tracking from agreement history	S
FR-20	Versioned read-only API with keys and metering	C

4 Source strategy

Rights come before reach. Inclusion of a site in this list is not authorisation to scrape or republish it. For every source, the terms and `robots.txt` are read and recorded in `data_sources` before it is switched on, and the answer is stored with the date it was checked.

4.1 Source tiers

Tier	What it is	How PFIE may use it
A	Public domain or open licence	Store, serve and redistribute
B	Free API under its own terms	Use within terms. Commercial use `CONFIRM` per provider
C	Official league and federation pages	Corroborate; store extracted facts only, not page content. Terms `CONFIRM` `ATTORNEY REVIEW`
D	Aggregator and media sites	Corroboration only, and only where terms and robots allow automated access. Default: off
X	Parallel Search	Finds pages. Never a source of truth

4.2 Source evaluation

Source	Competitions	Data	Method	Update rate	Access and reuse	Reliability (proposed)	Priority
OpenFootball (github.com/openfootball)	England four levels verified `VERIFIED`; other country repositories to check `CONFIRM`	Fixtures, results	Direct fetch of raw files, no Firecrawl credit	Varies by repository and season; freshness unknown `CONFIRM`	Data "dedicated to the public domain" `VERIFIED`. Tier A	High for structure, medium for freshness	1
football-data.org	Free plan lists 12 competitions `VERIFIED`	Fixtures, results (scores delayed on free)	API, 10 calls a minute `VERIFIED`	Frequent	Commercial use not stated on pricing page `CONFIRM`; ask the provider. Tier B	High	1
Premier League site	EPL	Fixtures, kick-offs	Firecrawl markdown, then Gemini	Updated by the league	Terms and robots `CONFIRM`. Tier C	High (official)	1
UEFA	UCL, UEL, UECL, Nations League, Euro	Fixtures, kick-offs	Firecrawl, Gemini	Updated by UEFA	Terms `CONFIRM`. Tier C	High (official)	1

Source	Competitions	Data	Method	Update rate	Access and reuse	Reliability (proposed)	Priority
Premier Soccer League (SA)	Betway Premiership, Motsepe Foundation Championship	Fixtures, results	Firecrawl, Gemini	Updated by the PSL	Terms CONFIRM. Tier C	High (official)	1
LaLiga, Bundesliga, Lega Serie A, Ligue 1	Own leagues	Fixtures	Firecrawl, Gemini	League-managed	Terms CONFIRM. Tier C	High (official)	2
CAF	CAF Champions League, Confederation Cup, AFCON	Fixtures	Firecrawl, Gemini	Irregular	Terms CONFIRM. Tier C	Medium	2
ESPN, BBC Sport	Broad	Fixtures, results	Only if terms permit CONFIRM	Frequent	Automated access is commonly restricted; verify. Tier D	Medium-high	3
Soccerway, WorldFootball.net	Broad	Fixtures, results	Only if terms permit CONFIRM	Frequent	Verify. Tier D	Medium	3
Parallel Search	Not a data source	URL discovery	API, $5 monthly credit VERIFIED	On demand	Card activation and overage billing VERIFIED; commercial terms CONFIRM. Tier X	n/a	Discovery only

Reliability is measured, not assumed. `data_sources.reliability` is recalculated from how often each source's observations agree with the final verified record.

4.3 Competition registry (51 targeted, all configurable)

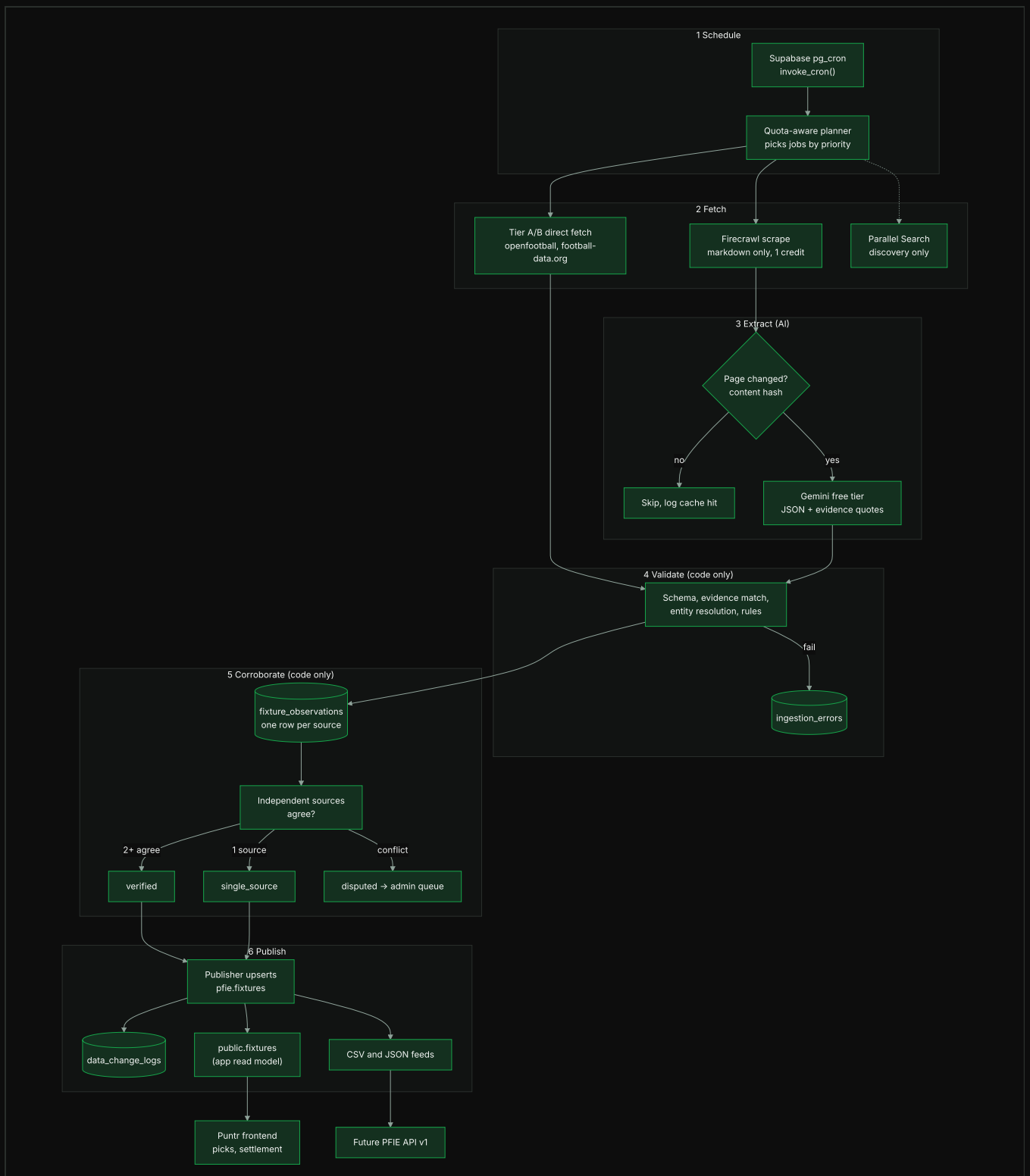
Each competition starts as `planned`. It becomes `active` only when at least one permitted source exists and the corroboration rule can be met. Coverage is never promised ahead of that.

Group	Competitions
England (6)	Premier League, Championship, League One, League Two, FA Cup, EFL Cup
Spain (3)	LaLiga, Segunda División, Copa del Rey
Germany (3)	Bundesliga, 2. Bundesliga, DFB-Pokal
Italy (3)	Serie A, Serie B, Coppa Italia

Group	Competitions
France (3)	Ligue 1, Ligue 2, Coupe de France
Europe (5)	UEFA Champions League, Europa League, Conference League, Nations League, Euro
Other Europe (7)	Eredivisie, Eerste Divisie, Primeira Liga, Scottish Premiership, Scottish Championship, Süper Lig, Belgian Pro League
South Africa (4)	Betway Premiership, Motsepe Foundation Championship, Nedbank Cup, MTN8
Africa (3)	CAF Champions League, CAF Confederation Cup, Africa Cup of Nations
International (3)	FIFA World Cup, CAF World Cup qualifiers, UEFA World Cup qualifiers
Other markets (11)	Saudi Pro League, MLS, Brazilian Série A, Argentine Primera División, Liga MX, J1 League, A-League, Copa América, Copa Libertadores, Copa Sudamericana, AFC Champions League

Priority 1 (daily): 6. Betway Premiership, Premier League, Champions League, Europa League, Conference League, Motsepe Foundation Championship. **Priority 2 (twice weekly): about 15. Priority 3 (weekly): the rest.**

5 System architecture



Ingestion lifecycle

5.1 Components and responsibilities

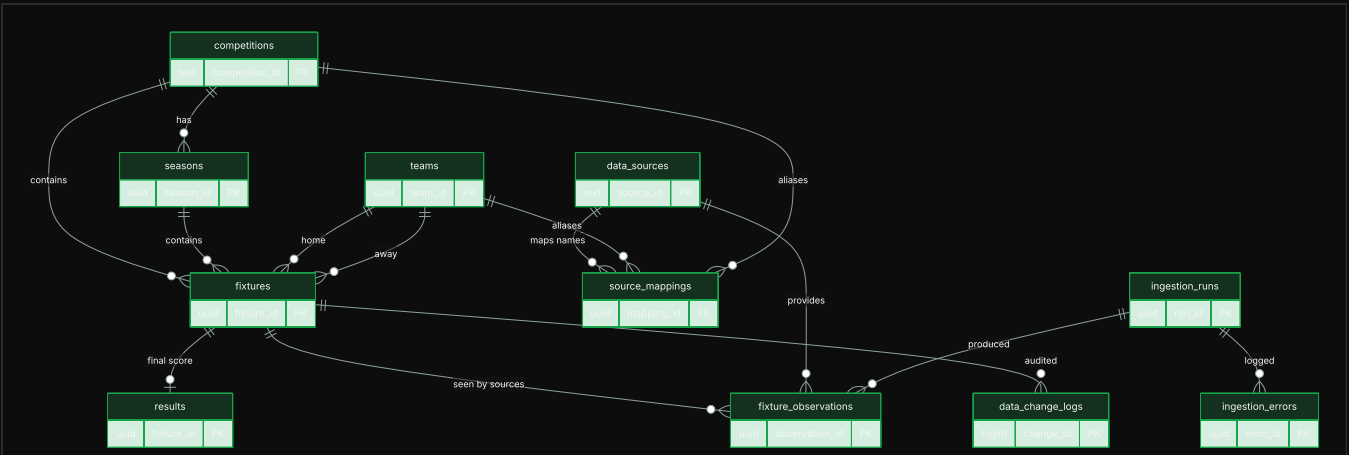
Component	Responsibility
Scheduler	Supabase `pg_cron` calls `public.invoke_cron(...)`, which calls PFIE server routes. This is the pattern already used by `sync-fixtures`, `sync-results` and `settle` in the repository VERIFIED. Vercel Cron is not used: Hobby allows one run a day VERIFIED.
Planner	Reads the quota ledger and the competition priority, then chooses which jobs run now. This is where the schedule shrinks when quota is short.
Fetchers	One adapter per source behind a common interface, extending the existing `SportsProvider` idea in `src/lib/sports/provider.ts`. Tier A and B sources use plain HTTP and cost no Firecrawl credit.
Firecrawl	Renders pages that need a browser and returns markdown. Server-side only. 1 credit a page VERIFIED.
Content hash cache	If a page's markdown is unchanged since the last run, Gemini is not called.
Extractor	Sends page text to Gemini with a fixed JSON schema. Returns candidate fixtures with an evidence quote for each field.
Validator	Pure code. Schema check, evidence check, football rules, entity resolution (section 7).
Observation store	Every validated sighting is saved as one row in `fixture_observations`, per source and run.
Corroborator	Pure code. Compares observations of the same fixture and sets `verified`, `single_source` or `disputed`.
Publisher	Upserts into `pfie.fixtures` and `pfie.results`, writes `data_change_logs`, and projects verified fixtures into the existing `public.fixtures` read model so picks and settlement keep working.
Exporter	CSV and JSON feeds from verified data.
Monitor	Quota ledger, run history, error rates, staleness alarms.
Admin queue	Screen in `/admin` for disputed fixtures and unresolved team names.

5.2 Repository assessment

The repository already provides: a `public.fixtures` table (text `competition`, `home`, `away`, `kickoff_at`, `status`, scores, `provider`, `provider_ref`, unique on provider and reference), a provider interface, a competition registry file, an API-Football adapter whose account is currently suspended, cron routes for fixture and result sync, a settlement engine that grades picks from finished fixtures, `pg_cron` with an `invoke_cron` helper, and Row Level Security on every table.

Decision: do not break settlement. PFIE writes to its own `pfie` schema and publishes to `public.fixtures` with `provider = 'pfie'` and the canonical fixture key as `provider_ref`. The app and `settle.server.ts` keep reading one table. Grading only ever runs on results PFIE has marked `verified`.

6 Data model



Data model

All tables live in schema `pfie`, have Row Level Security on, and grant nothing to `anon` or `authenticated`. Only server code using the service role writes. The app reads through `public.fixtures` and through server routes.

Canonical IDs. `competition_id` is a stable slug (`eng-premier-league`). `team_id` is a UUID. Source names never appear in canonical tables: they live in `source_mappings`. The natural key of a fixture is `(competition_id, season, round, home_team_id, away_team_id, leg)`. `fixture_id` is derived from it, so a second ingestion of the same fixture updates the row and cannot duplicate it (idempotent ingestion).

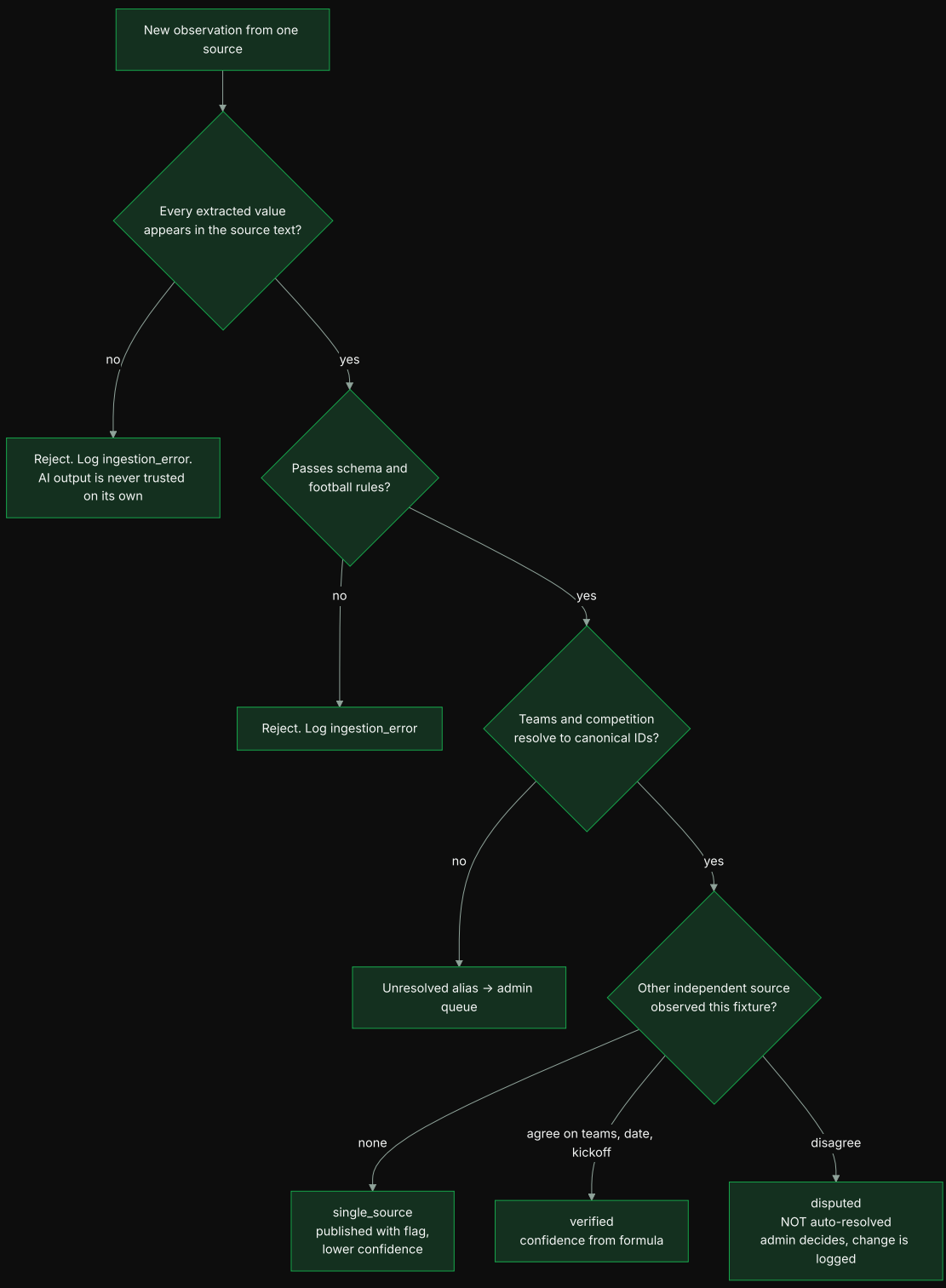
Time. `kickoff_utc timestamptz` is the only stored time. `kickoff_local` is computed in `Africa/Johannesburg` at read time using the existing helpers in `src/lib/sast.ts`.

Table	Key fields	Constraints and indexes
`competitions`	`competition_id` PK, `name`, `country`, `tier`, `priority`, `season_rule`, `status` (planned, active, paused)	unique `name` per country
`teams`	`team_id` PK, `canonical_name`, `country`, `short_name`	unique (`canonical_name`, `country`)
`seasons`	`season_id` PK, `competition_id` FK, `label`, `starts_on`, `ends_on`	unique (`competition_id`, `label`)
`fixtures`	`fixture_id` PK, `competition_id`, `season_id`, `round`, `leg`, `home_team_id`, `away_team_id`, `kickoff_utc`, `venue`, `match_status`, `verification_status`, `confidence_score`, `source_count`, `last_verified_at`, `last_updated_at`	unique natural key; index (`kickoff_utc`), (`match_status`, `kickoff_utc`); check home is not away
`results`	`fixture_id` PK/FK, `home_score`, `away_score`, `verified_at`, `source_count`	check scores are 0 or more; only written when `finished`
`data_sources`	`source_id` PK, `domain`, `tier`, `terms_checked_on`, `robots_ok`, `reliability`, `enabled`	unique `domain`

Table	Key fields	Constraints and indexes
`source_mappings`	`mapping_id` PK, `source_id`, `entity_type`, `source_name`, `team_id` or `competition_id`	unique (`source_id`, `entity_type`, `source_name`)
`fixture_observations`	`observation_id` PK, `run_id`, `source_id`, `fixture_id` (nullable until resolved), `raw_extract` jsonb, `evidence` jsonb, `content_hash`, `observed_at`	index (`fixture_id`, `source_id`)
`ingestion_runs`	`run_id` PK, `job`, `started_at`, `finished_at`, `status`, `calls_firecrawl`, `calls_gemini`, `calls_parallel`	index (`job`, `started_at`)
`ingestion_errors`	`error_id` PK, `run_id`, `source_id`, `stage`, `code`, `detail`	index (`run_id`)
`data_change_logs`	`change_id` PK, `fixture_id`, `field`, `old_value`, `new_value`, `reason`, `actor`, `changed_at`	append only (no update or delete grants)

7 AI extraction workflow

The rule. Gemini reads a page and proposes facts. Code checks them. Several independent sources agree or they do not. No step lets a model choose between conflicting versions.



Truth model

7.1 Steps

1. **Fetch.** Firecrawl returns markdown, or a Tier A/B fetcher returns structured data. A content hash skips unchanged pages.
2. **Extract.** One Gemini call per page, temperature 0, response constrained to a JSON schema. For every field the model must return the exact text it read it from (an evidence quote). Several fixtures per call keeps request volume low.
3. **Evidence check (code).** Each extracted value must be found in the page text: team names as text, the date and time as a parseable string in the quoted line. If not, the record is rejected and logged. This is what stops a model from inventing a fixture.
4. **Schema and rules (code).** Valid types and ranges, kick-off inside the season window, home is not away, both teams belong to the competition, scores only on finished matches, allowed status changes only (a finished match never returns to scheduled).
5. **Entity resolution (code).** Names are matched through `source_mappings`. An exact alias match resolves. A near match goes to the admin queue as a proposed alias. Nothing is guessed.
6. **Observe.** The validated record is saved as an observation for that source and run.
7. **Corroborate (code).** Two or more independent sources that agree on teams, date and kick-off within five minutes make the fixture `verified`. One source makes it `single_source`. Disagreement makes it `disputed` and it goes to the admin queue. Independent means a different domain and a different upstream: two sites quoting the same league page count once.
8. **Publish.** Verified and single-source fixtures are written with their state. Disputed ones are held.

7.2 Confidence score (proposed, fixed formula, tunable)

Situation	Score
Two or more independent sources agree	0.90
One Tier A or Tier C (official) source	0.75
One other source	0.50
Alias matched fuzzily and approved	minus 0.05
Kick-off changed since last run, not yet re-confirmed	minus 0.15
Source's rolling agreement rate below 90%	minus 0.10
Disputed	recorded as 0 and held

The score is arithmetic on recorded facts. The model never reports its own confidence and the app never reads one from it.

7.3 Integrity rules for a pick-locking product PROPOSED

Puntr locks picks at kick-off, so a wrong kick-off time damages trust. Proposed, owner to decide:

- A pick can open on a `verified` fixture, or a `single_source` fixture from an official source. A fixture with only low-confidence sources is shown as "time to be confirmed" and takes no picks.
- If sources disagree on kick-off, the **earliest credible time** is used for locking until an admin resolves it.

- A kick-off moved later does not reopen picks that already locked.
- A result is only graded when it is `verified` (two sources, or an official Tier A/C source plus no conflict). Anything else waits in the admin queue.

7.4 Safety of the extraction itself

- Page text is **untrusted input**. The model is given no tools and no instructions to act, its output is schema-only, and the validator ignores free text. A page that tries to instruct the model cannot change what is stored.
- **No personal data is ever sent to Gemini or Firecrawl.** Only public fixture pages are fetched. Cookies, forms and account pages are never requested.
- Gemini rate limits for the free tier are not published on the pricing or rate-limit pages and must be read from Google AI Studio at implementation `CONFIRM`. The code reads the limit from configuration and never assumes it.
- Current Google guidance is to use a current Flash or Flash-Lite model for new projects `VERIFIED`. The model name is a configuration value.

8 Zero-cost operating model

Target: R0 of additional cost. The approved providers are Firecrawl, Gemini, Parallel, Supabase, Vercel and GitHub, on free tiers only. Claude is a development assistant only: production never calls a paid model.

8.1 Provider-by-provider free tier (verified 5 November 2026)

Provider	Free allowance	Rate limits	Terms to settle
Firecrawl	1,000 credits a month, no card VERIFIED. Scrape 1 credit, search 2 credits per 10 results, JSON mode +4 VERIFIED	10 a minute on scrape, map and search; 2 a minute on crawl; 2 concurrent browsers VERIFIED	Page does not state commercial-use terms for Free; no Data Processing Agreement on Free VERIFIED. CONFIRM ATTORNEY REVIEW
Google Gemini	Free tier exists; do not rely on any figure here CONFIRM	Per project, read from AI Studio; daily quotas reset at midnight Pacific VERIFIED	Free-tier content is used to improve Google products VERIFIED. Public data only
Parallel Search	$5 monthly credit per organisation, about 5,000 searches, expires monthly VERIFIED	Search API 600 a minute VERIFIED	Card activates the credit and overage is billed to it VERIFIED. Commercial terms not stated CONFIRM
Supabase	500 MB database, 5 GB egress, 500,000 edge invocations, 50,000 monthly active users VERIFIED	n/a	Projects pause after one week of inactivity; no automatic backups on Free VERIFIED
Vercel	Hobby: 100 cron jobs, once a day, run within the hour (±59 minutes) VERIFIED	n/a	Hobby is non-commercial only; affiliate links and advertising count as commercial VERIFIED
GitHub	Existing repository and free Actions minutes CONFIRM	n/a	None known for this use

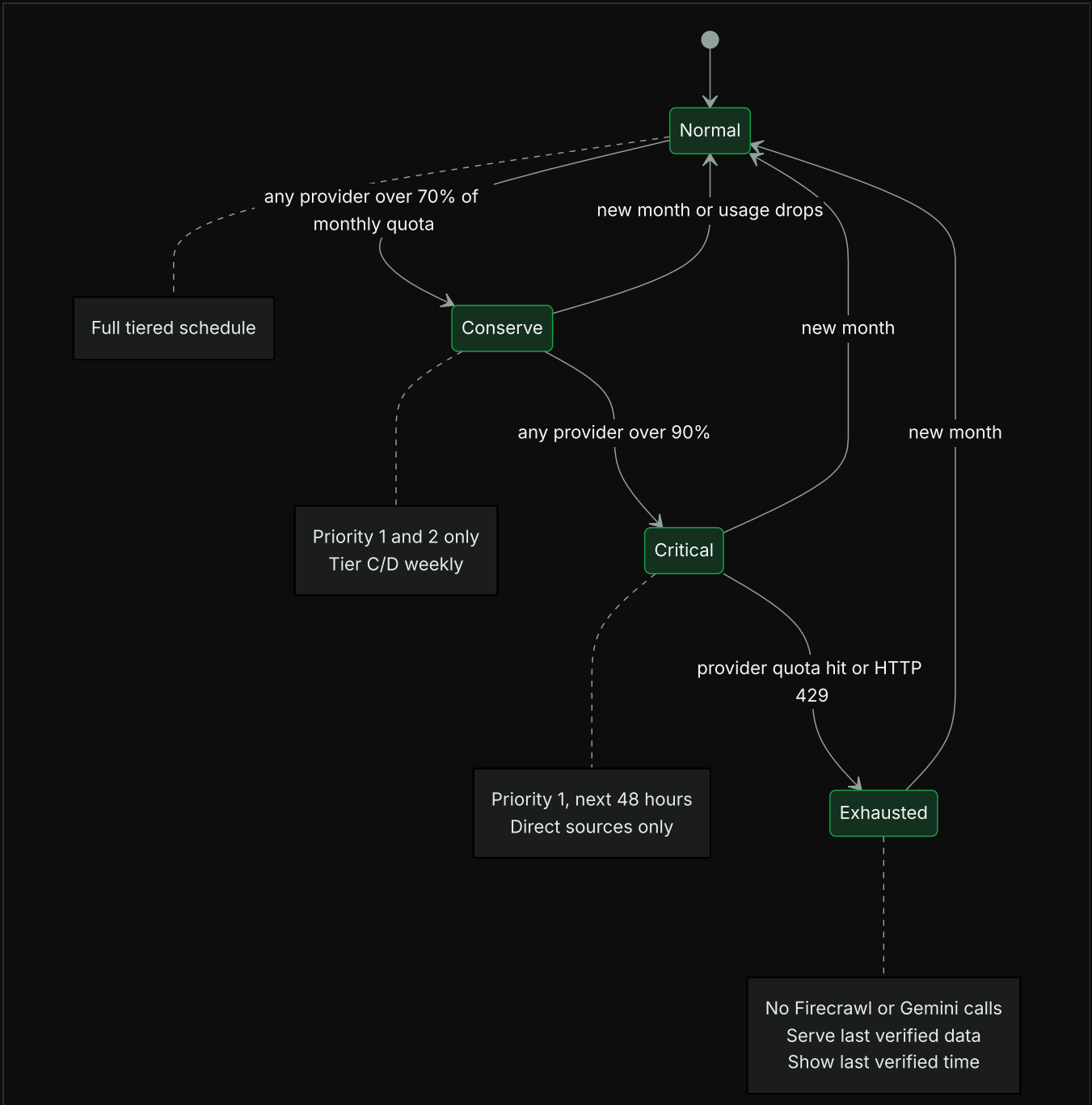
8.2 Volume budget for Firecrawl (1,000 credits a month)

Use	Calculation	Credits a month
Priority 1 fixtures pages, daily	6 competitions × 30 days	180
Priority 2 pages, twice weekly	15 × 2 × 4.3	129
Priority 3 pages, weekly	30 × 4.3	129
Match-day result verification	about 7 a day	200
Parallel replaces Firecrawl search for discovery	0	0
Planned use		**638**
Reserve for retries and ad hoc fixes	about 25%	250
Total		**888 of 1,000**

The brief's target schedule does not fit free tiers. Refreshing 50 competitions every six hours is about 6,000 credits a month. The schedule below is what the budget supports, and the planner reduces it further when quota tightens.

Job	Target in brief	Free-tier schedule PROPOSED
Full upcoming collection	every 6 hours	Daily for Priority 1; twice weekly Priority 2; weekly Priority 3. Tier A/B sources cost no credits and can run more often
Verification	every 3 hours	Re-observe from a second source only when a fixture is within 7 days or unverified
Next 48 hours	hourly	Every 3 hours for Priority 1 only, using direct sources first
Postponements	every 2 hours	Folded into the 48-hour job; weekend mornings get an extra run
Results	every 30 minutes	Every 30 minutes only for matches finished within the last 6 hours, direct sources first
Historical reconciliation	daily	Daily, database only, no external calls
Integrity audit	weekly	Weekly, database only

8.3 Quota monitoring, graceful degradation and exhaustion



Quota modes

- A **quota ledger** records every call by provider, job and day. The planner reads it before every job.
- Free allowances are **configuration**, not constants, so a changed provider limit is a settings change, not a release.
- Requests are **deduplicated** by URL and content hash within a run and across runs.
- In **Exhausted** mode nothing breaks for members. They see the last verified data with its "last verified" time, and the admin sees an alert.

- **No-cost fallbacks:** Tier A and B sources, the previous verified data, and manual CSV import from the admin screen.

8.4 Cost-related decisions for the owner

1. **Vercel plan.** If the project is on Hobby, the commercial-use restriction needs a decision. Moving to a paid plan costs money and conflicts with R0. PFIE does not change this, and does not depend on Vercel Cron.
2. **Parallel.** Use it only if its card can be one with no ability to be charged (for example a prepaid card), or skip it. Discovery can be done by hand for 50 competitions.
3. **Firecrawl and Parallel commercial terms.** Ask in writing before relying on either in a product.

9 Security and reliability

Area	Design PROPOSED
Secrets	Firecrawl, Gemini, Parallel and Supabase service keys exist only in server environment variables. They are never in client code, the repository or logs
Database access	Schema `pfie` with Row Level Security on every table, no grants to `anon` or `authenticated`, service role only. The app reads `public.fixtures`
Input validation	Every extraction passes a schema and rules check. Admin inputs use the same schemas. URLs fetched come only from the `data_sources` allow-list
Rate limiting	Per-provider limiter inside the fetchers, set from configuration. Cron routes require the existing cron secret check (`cron-auth.server.ts`). Public feeds are cached and rate limited
Logging	`ingestion_runs` and `ingestion_errors` for every run. Logs never contain secrets or personal data
Errors and retries	At most two retries with backoff, then the job stops for that source and logs the error. No infinite loops, no retry storms
Idempotent ingestion	Natural key and content hash. Re-running a job changes nothing if nothing changed
Provenance	Every field traces to an observation, a source, a URL and a time. `data_change_logs` is append only
Backup and recovery	Supabase Free has no automatic backups VERIFIED. Proposed: a nightly GitHub Actions job calls an authenticated export route and commits CSV snapshots of the PFIE tables to a private repository (facts only, no personal data). Target recovery point 24 hours. Restore is an import script tested before launch
Inactivity pause	Supabase pauses Free projects after a week without activity VERIFIED; scheduled jobs keep PFIE active, and a weekly check confirms it

10 Development roadmap

Two dates are kept apart. Technical completion is **Monday 9 November 2026**. The public launch is **Tuesday 1 December 2026**. The 22 days between them are for coverage, quality and monitoring, not for finishing the platform.

Phase	When	Work	Exit
0. Founding acquisition	Now, before PFIE	Early-access landing page `/founding`, sitewide Founding bar, home hero and header and footer copy, sign-up forms already live at `/join`. Social assets and reel prepared. Built; pull request #42 waits for the owner to merge it	Merged and live on puntr.co.za
1. Repository assessment and foundation	Fri 6 Nov	Migration for schema `pfie`, RLS, seed registry of 51 competitions, `data_sources` and quota ledger	Migration applied, types regenerated
2. Tier A and B integrations	Fri 6 to Sat 7 Nov	OpenFootball importer, football-data.org adapter, entity resolution and alias tables	England fixtures flowing at zero credits
3. Extraction and normalisation	Sat 7 to Sun 8 Nov	Firecrawl fetcher, content hash cache, Gemini extractor with evidence quotes, validator	Priority 1 competitions extracting
4. Corroboration and publishing	Sun 8 Nov	Observation store, corroborator, confidence formula, publisher into `public.fixtures`, admin queue	Verified, single-source and disputed states visible
5. Exports and Puntr integration	Sun 8 to Mon 9 Nov	CSV and JSON feeds, `last verified` shown in the app, backup job	Feeds download; settlement still passes its tests
6. QA and deployment	Mon 9 Nov	Data checks against official sources for Priority 1, failure and exhaustion drills, deploy	Technical completion
7. Coverage and monitoring	10 to 30 Nov	Switch competitions to `active` one by one, tune thresholds, watch quota and errors	Launch readiness checklist green
8. Launch and post-launch	From 1 Dec	Daily monitoring, weekly integrity audit, source terms re-check	Stable first month
9. API design	Q1 2027	Section 13 stage gates	Decision to build

11 Launch readiness checklist

The detailed checklist, with owners and checkboxes, is in `PFIE-Implementation-Checklist.md`. The launch gates are:

- ☐ `/founding`, the sitewide bar and the new hero are live, and sign-ups are being counted
- ☐ Priority 1 competitions are `active` with at least two sources each, or an official source plus a stored terms check
- ☐ No fixture inside the next 7 days is `disputed` and unreviewed
- ☐ Every record shows source and last verified time
- ☐ Settlement grades only `verified` results
- ☐ Quota ledger below 70% on 29 November
- ☐ Backup job has run and a restore has been tested
- ☐ Terms and robots checks are recorded for every enabled source
- ☐ Legal copy is aligned: `/terms`, `/privacy`, `/cookies`, `/affiliate-disclosure`, `/refunds` and the legal changelog ATTORNEY REVIEW
- ☐ No page claims live scores, odds or complete coverage

12 Risks and mitigations

Risk	Likelihood	Impact	Mitigation
A wrong fixture or kick-off reaches members	Medium	High	Evidence check, corroboration, earliest-time lock rule, admin queue, audit log
Free tier quota runs out mid-month	High	Medium	Tiered schedule, 25% reserve, quota modes, direct sources, manual CSV
A provider changes or withdraws its free tier	Medium	Medium	Provider adapters, configuration limits, tier A/B fallbacks
Scraping is not permitted for a site	Medium	High	Terms and robots recorded before enabling; official and public-domain sources first; legal review
Fixture data is protected by database rights	Medium	High	Store facts only; do not republish source pages; do not resell third-party facts without a licence ATTORNEY REVIEW
Free-tier Gemini content use	Certain	Medium	Public pages only; no personal data; no private prompts
Parallel or Vercel charges appear	Medium	Low to medium	Card without charge ability; confirm Vercel plan; alert on usage
Prompt injection in a scraped page	Low	Medium	No model tools, schema-only output, validator ignores free text
Four-day schedule to 9 November slips	High	Medium	Core slice defined in section 1; coverage continues to 30 November
Supabase pause or data loss	Low	High	Scheduled activity, nightly export, tested restore
Matches postponed after kick-off time changes	High	Medium	Frequent 48-hour checks; status rules; picks policy in 7.3
Founding sign-ups stall while engineering focuses on PFIE	Medium	High	Phase 0 first; campaign assets ready; weekly sign-up review

13 Future development opportunities

13.1 From PFIE to a data API PROPOSED

PFIE is designed so that a public API is an addition, not a rewrite: stable canonical IDs, provenance on every record, versioned shapes and append-only history.

Stage	Description	Gate to pass first
S0	Internal feeds: CSV and JSON for Puntr itself	Done in this blueprint
S1	Read-only `/v1` for Puntr-original data only: aggregated consensus and graded track records, with no personal data	Counsel review of data protection (POPIA) and terms of use
S2	API keys, tenants, rate limits, usage metering, a developer page, response fields for source and verification	Abuse controls and a support process
S3	Third-party facts added only from licensed or public-domain sources	Written licences; Tier A coverage review
S4	Prediction data product: calibrated model outputs with provenance, for partners	Evidence of track-record quality; marketing claims reviewed so that nothing implies profit ATTORNEY REVIEW

13.2 What can be proprietary

- **Can be Puntr's own:** the verification metadata, the canonical ID system, community consensus, graded pick history, derived indicators and model outputs.
- **Cannot be assumed to be Puntr's:** raw fixture and result data taken from third-party sites. These are facts, but their collection can carry database rights, and sites' terms apply. The API's value should rest on the first list.

13.3 Other opportunities

- Women's football, youth and lower South African leagues as sources allow.
- Venue, referee and weather enrichment from open sources.
- Webhook notifications for fixture changes (to partner platforms and Puntr's own alerts).
- Multi-sport expansion on the same registry and observation model, following the `/roadmap` schedule.
- Public verification pages showing the provenance of any record, supporting Puntr's "prove it" positioning.

Appendix A. Fixture schema

Field	Type	Notes
`fixture_id`	uuid	derived from the natural key
`competition_id`	text	canonical slug
`competition_name`	text	display name
`season`	text	for example `2026-27`
`round`	text	matchweek or stage
`home_team_id`, `home_team_name`	uuid, text	canonical
`away_team_id`, `away_team_name`	uuid, text	canonical
`match_date`	date	local match date (Africa/Johannesburg)
`kickoff_utc`	timestamptz	stored time
`kickoff_local`	timestamptz	computed for display, `Africa/Johannesburg`
`venue`	text	optional
`match_status`	enum	scheduled, live, finished, postponed, abandoned, cancelled
`home_score`, `away_score`	integer	null unless finished
`source_id`, `source_url`	text	the source behind the published values
`confidence_score`	numeric(3,2)	from the formula in 7.2
`last_verified_at`, `last_updated_at`	timestamptz	provenance

Example CSV record (illustrative values, not a real fixture):

```
fixture_id,competition_id,competition_name,season,round,home_team_id,home_team_name,away_team_id,away_team_name,match_date,kickoff_utc,kickoff_local,venue,match_status,home_score,away_score,source_id,source_url,confidence_score,last_verified_at,last_updated_at
7f0c2a1e-0000-4000-8000-000000000001,rsa-betway-premiership,Betway Premiership,2026-27,Round 10,3b1d9c10-0000-4000-8000-0000000000a1,Example FC,3b1d9c10-0000-4000-8000-0000000000a2,Sample United,2026-11-21,2026-11-21T15:30:00Z,2026-11-21T17:30:00+02:00,Example Stadium,scheduled,,,psl-za,https://example.org/fixtures,0.90,2026-11-19T06:00:00Z,2026-11-19T06:00:00Z
```

Appendix B. Editable diagram sources

Mermaid source files are in `docs/pfie/diagrams/` : `ingestion-lifecycle.mmd` , `data-model.mmd` , `truth-model.mmd` , `quota-modes.mmd` , `product-layers.mmd` , with the shared theme in `mmd.json` . Re-render with `mmdc -i <file>.mmd -o <file>.svg -c mmd.json -b '#0D0D0D'` .

Appendix C. Open decisions for the owner

#	Decision	Recommended
D1	Vercel plan and the commercial-use restriction	Confirm the current plan; decide before launch
D2	Use Parallel at all	Only with a card that cannot be charged, or skip
D3	Written confirmation of commercial use for Firecrawl and football-data.org	Ask this week
D4	Pick-lock rules in 7.3	Approve as proposed
D5	Scope of 9 November	Core slice in section 1
D6	Where S1 of the API sits on the roadmap	After launch, subject to counsel

Appendix D. Sources checked on 5 November 2026

- Firecrawl pricing and rate limits: https://www.firecrawl.dev/pricing and https://docs.firecrawl.dev/rate-limits
- Gemini pricing, rate limits and models: https://ai.google.dev/gemini-api/docs/pricing, https://ai.google.dev/gemini-api/docs/rate-limits, https://ai.google.dev/gemini-api/docs/models
- Parallel free tier and pricing: https://parallel.ai/blog/free-tier-parallel and https://parallel.ai/pricing
- Supabase pricing: https://supabase.com/pricing
- Vercel cron limits and fair use: https://vercel.com/docs/cron-jobs/usage-and-pricing and https://vercel.com/docs/limits/fair-use-guidelines
- OpenFootball England data: https://github.com/openfootball/england
- football-data.org pricing: https://www.football-data.org/pricing
- Repository: `supabase/migrations/20261029071500_mvp_core.sql`, `src/lib/sports/provider.ts`, `supabase/migrations/20261030070000_cron_schedule.sql`